Cost-Tilted List Decoding under General Positive Weights

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Abstract

We consider an arbitrary block-length-n code (e, d) over a discrete memoryless channel $W\colon \mathcal{X} \to \mathcal{P}(\mathcal{Y})$ with M equiprobable messages. Given any strictly positive "tilt" function

$$w\colon \{1, \dots, M\} \times \mathcal{Y}^n \to (0, \infty), \quad w_{\min} = \min_{m, y^n} w(m, y^n) > 0,$$

we define

$$p(m, y^n) = \frac{1}{M} W^n\big(y^n \mid e(m)\big), \quad \mathcal{Z}_{\text{tot}} = \sum_{m=1}^{M} \sum_{y^n} w(m, y^n)\, p(m, y^n),$$

and the tilted joint distribution

$$p'(m, y^n) = \frac{w(m, y^n)\, p(m, y^n)}{\mathcal{Z}_{\text{tot}}}.$$

Under p', let

$$Z = \mathbf{1}\{m \notin \mathcal{L}(Y^n)\}, \quad P = \Pr_{p'}[Z = 1],$$

where $\mathcal{L}(y^n) \subseteq \{1, \dots, M\}$ is any list satisfying $|\mathcal{L}(y^n)| \le L$. Define the tilted conditional entropy

$$H = H_{p'}\big((M, Y^n) \mid Z = 1\big), \quad \tau = 2^{-H}, \quad L = \lfloor 1/\tau \rfloor.$$

Then by choosing

$$\mathcal{L}(y^n) = \{\, m : p'(m \mid y^n) \ge \tau \}, \quad p'(m \mid y^n) = \frac{p'(m, y^n)}{\sum_{m'} p'(m', y^n)},$$

one obtains a valid list-L decoder whose error probability under the true law p satisfies

$$P_e^{(L)} \;\le\; \frac{\mathcal{Z}_{\text{tot}}}{w_{\min}}\, P.$$

When $w \equiv 1$ this recovers the standard single-shot converse for $L = 1$, and under mild moment or concentration conditions one shows $\mathcal{Z}_{\text{tot}}/w_{\min} = e^{o(n)}$, so that the dominant exponential rate is governed purely by the tilted error probability P.

1 Preliminaries

We fix a block-length n, finite alphabets $\mathcal{X}, \mathcal{Y}$, and a DMC

$$W^n(y^n \mid x^n) = \prod_{i=1}^{n} W(y_i \mid x_i).$$

We use an M-message code (e, d) with $e\colon \{1, \dots, M\} \to \mathcal{X}^n$ and $d\colon \mathcal{Y}^n \to 2^{\{1, \dots, M\}}$, and a strictly positive tilt

$$w\colon \{1, \dots, M\} \times \mathcal{Y}^n \to (0, \infty), \quad w_{\min} = \min_{m, y^n} w(m, y^n) > 0.$$

Table 1: Key notation

Symbol	Description
M	Number of messages/codewords
$e\colon \{1,\dots,M\} \to \mathcal{X}^n$	Encoder map
$d\colon \mathcal{Y}^n \to 2^{\{1,\dots,M\}}$	List-decoder map
$W(y \mid x)$	DMC transition probability
$w(m, y^n)$	Arbitrary positive "tilt" weight function
$w_{\min}$	$\min_{m,y^n} w(m, y^n)$
$p(m, y^n)$	Joint law $= \frac{1}{M} W^n(y^n \mid e(m))$
$\mathcal{Z}_{\text{tot}}$	Total normalizer $= \sum_m \sum_{y^n} w(m, y^n)\, p(m, y^n)$
$p'(m, y^n)$	Tilted distribution $\propto w(m, y^n)\, p(m, y^n)$
Z	Error indicator $\mathbf{1}\{m \notin \mathcal{L}(y^n)\}$
$P = \Pr_{p'}[Z = 1]$	Tilted error probability under p'
$H = H_{p'}((M, Y^n) \mid Z = 1)$	Tilted conditional entropy
$\tau = 2^{-H}$	Decision threshold for list membership
$L = \lfloor 1/\tau \rfloor$	Resulting list size

All logs are base-2, and we write

$$D_{\text{KL}}(P\|Q) = \sum_x P(x) \log_2 \frac{P(x)}{Q(x)}.$$

Under this code define the true joint

$$p(m, y^n) = \frac{1}{M} W^n(y^n \mid e(m)),$$

the normalizer

$$\mathcal{Z}_{\text{tot}} = \sum_{m,y^n} w(m, y^n)\, p(m, y^n),$$

and the tilted joint

$$p'(m, y^n) = \frac{w(m, y^n) p(m, y^n)}{\mathcal{Z}_{\text{tot}}}.$$

2 Related Work

Our unified single-shot list-decoding converse under arbitrary positive cost tilts generalizes and bridges several strands of converse analyses:

- **Finite-blocklength and meta-converse results.**
  - If $w \equiv 1$ and $L = 1$, Proposition 2 reduces exactly to the single-message finite-blocklength converse of [1].
  - The moderate-deviations limits of [2] follow by letting $R \to C$, $n \to \infty$.
  - The channel-with-state meta-converse of [3] captures message-dependent tests via a minimax formulation analogous to our tilted hypothesis test.
- β–β **hypothesis-testing bounds and cost constraints.**
  - Specializing the information-spectrum change-of-measure of [4] to two hypotheses ($M = 2$, $L = 1$) and a message-independent tilt recovers the classical β–β trade-off.

- [5] extends the β–β second-order analysis to channels under arbitrary input or state constraints.
- [7] develop joint source–channel converses via Neyman–Pearson tests with message-dependent weights, closely paralleling our tilted-test formulation.

- **Source-coding dispersion.**
  - [6] derives tight finite-blocklength achievability and converse bounds for fixed-length lossy compression, showing

$$R_n(d,\epsilon) \approx R(d) + \sqrt{\tfrac{V(d)}{n}}\, Q^{-1}(\epsilon),$$

    where $V(d)$ is the rate-dispersion of the source.

- **Classical error-exponent and unequal-protection approaches.**
  - [8] tilt the channel law by a cost; our message-dependent tilt extends these to the one-shot/list setting.
  - [9] provides finite-blocklength bounds for multiple message-classes, akin to arbitrary tilt functions.

- **Information-spectrum and foundational methods.**
  - [10] uses a hypothesis-testing change of measure, serving as the asymptotic precursor to our finite-n analysis.
  - [11] and [12] offer foundational perspectives on message-dependent hypothesis tests.

3 Main Result

Assume the weight family $w(m, y^n)$ satisfies either the moment-control of Lemma 4 (with $\ln w_{\min} = -o(n)$) or—under the memoryless-tilt assumption—the concentration of Lemma 5.

Define

$$p(m, y^n) = \frac{1}{M}\, W^n\big(y^n \mid e(m)\big), \qquad \mathcal{Z}_{\text{tot}} = \sum_{m=1}^{M} \sum_{y^n} w(m, y^n)\, p(m, y^n),$$

and the tilted joint

$$p'(m, y^n) = \frac{w(m, y^n)\, p(m, y^n)}{\mathcal{Z}_{\text{tot}}}, \qquad \sum_{m, y^n} p'(m, y^n) = 1.$$

To select our list-size threshold, define the function

$$f(\tau) = H_{p'}\big((M, Y^n) \mid p'(M \mid Y^n) < \tau\big) + \log_2 \tau, \qquad \tau \in (0, 1].$$

Lemma 1 (Existence of threshold)**.** *The map $\tau \mapsto f(\tau)$ is continuous on $(0, 1]$. Moreover:*

- *As $\tau \to 0^+$, the term $\log_2 \tau \to -\infty$, so $f(\tau) \to -\infty$.*
- *At $\tau = 1$, the condition $p'(M \mid Y^n) < 1$ holds almost surely, hence*

$$f(1) = H_{p'}(M, Y^n) + \log_2 1 = H_{p'}(M, Y^n) \;\geq\; 0.$$

By the intermediate-value theorem, there must exist at least one $\tau^* \in (0, 1]$ such that

$$f(\tau^*) = 0, \quad H_{p'}\big((M, Y^n) \mid p'(M \mid Y^n) < \tau^*\big) = -\log_2 \tau^*.$$

Fix any such root τ^*, and set

$$H \;=\; -\log_2 \tau^*, \quad L \;=\; \left\lfloor \tfrac{1}{\tau^*} \right\rfloor,$$

$$\mathcal{L}(y^n) = \{\, m : p'(m \mid y^n) \geq \tau^* \}, \quad p'(m \mid y^n) = \frac{p'(m, y^n)}{\sum_{m'} p'(m', y^n)},$$

$$Z = \mathbf{1}\{\, m \notin \mathcal{L}(y^n)\}, \quad P = \Pr_{p'}[Z = 1].$$

By construction $|\mathcal{L}(y^n)| \leq 1/\tau^* \leq L$ for every y^n.

Proposition 2. *Let $\tilde{d}$ be any list-L decoder satisfying*

$$\mathcal{L}(y^n) \;\subseteq\; \tilde{d}(y^n) \quad \forall\, y^n.$$

Then its error probability under the true law p obeys

$$P_e^{(L)} = \sum_{y^n} \sum_{m \notin \tilde{d}(y^n)} p(m, y^n) \;\leq\; \frac{\mathcal{Z}_{\text{tot}}}{w_{\min}}\, P.$$

Proof. 1. (*List-size bound.*) Since $\sum_m p'(m \mid y^n) = 1$ and each $m \in \mathcal{L}(y^n)$ has $p'(m \mid y^n) \geq \tau^*$,

$$1 \;\geq\; \sum_{m \in \mathcal{L}(y^n)} p'(m \mid y^n) \;\geq\; |\mathcal{L}(y^n)|\, \tau^* \;\implies\; |\mathcal{L}(y^n)| \leq \frac{1}{\tau^*} \leq L.$$

2. (*Change of measure.*) For every (m, y^n),

$$p(m, y^n) = \frac{\mathcal{Z}_{\text{tot}}}{w(m, y^n)}\, p'(m, y^n) \;\leq\; \frac{\mathcal{Z}_{\text{tot}}}{w_{\min}}\, p'(m, y^n).$$

Summing over $m \notin \mathcal{L}(y^n)$ gives

$$\sum_{m \notin \mathcal{L}(y^n)} p(m, y^n) \;\leq\; \frac{\mathcal{Z}_{\text{tot}}}{w_{\min}} \sum_{m \notin \mathcal{L}(y^n)} p'(m, y^n).$$

3. (*Error-probability bound.*) Since $\tilde{d}(y^n) \supseteq \mathcal{L}(y^n)$,

$$P_e^{(L)} = \sum_{y^n} \sum_{m \notin \tilde{d}(y^n)} p(m, y^n) \;\leq\; \sum_{y^n} \sum_{m \notin \mathcal{L}(y^n)} p(m, y^n) \;\leq\; \frac{\mathcal{Z}_{\text{tot}}}{w_{\min}}\, P.$$

□

Corollary 3. *Specializing to $w \equiv 1$ and $L = 1$ recovers the standard single-shot finite-blocklength converse of [1].*

4 Design Example: BSC($\delta = 0.1$)

We illustrate how a small cost tilt can drastically reduce blocklength on a binary symmetric channel (BSC) with crossover probability $\delta = 0.1$. Fix a design rate R_{spec} and target error probability $P_e = 10^{-3}$. For each scalar tilt parameter $\alpha > 1$, proceed as follows:

1. **Define the per-bit tilt weights.** We write

$$w(m, y^n) \;=\; \prod_{i=1}^{n} w_1\big(m, y_i\big), \quad m \in \{1,2\},\; y_i \in \{0,1\},$$

where

$$w_1(1,1) = \alpha, \quad w_1(1,0) = 1, \qquad w_1(2,1) = 1, \quad w_1(2,0) = \alpha.$$

Equivalently,

$$w(m, y^n) = \begin{cases} \alpha^{\|y^n\|_1}, & m = 1, \\ \alpha^{\,n-\|y^n\|_1}, & m = 2. \end{cases}$$

Note that $w_{\min} = \min_{m,y^n} w(m, y^n) = 1$.

2. **Induced single-bit tilted channel.** Under the original code one has

$$p(m) = \tfrac{1}{2}, \qquad p(y^n \mid m) = \delta^{d_H(y^n, e(m))}(1-\delta)^{n-d_H(y^n, e(m))}.$$

Hence under the tilted joint each coordinate y_i is still independent, and

$$p'\big(y_i = 1 \mid m = 1\big) = \frac{\alpha\,\delta}{\alpha\,\delta + (1-\delta)}, \quad p'\big(y_i = 1 \mid m = 2\big) = \frac{1-\delta}{(1-\delta) + \alpha\,\delta}.$$

(shorthand: $p'(1 \mid 1) \equiv p'(y_i = 1 \mid m = 1)$ and $p'(1 \mid 2) \equiv p'(y_i = 1 \mid m = 2)$.)

Let

$$p_Y(1) = \tfrac{1}{2}\Big[p'\big(y_i = 1 \mid m = 1\big) + p'\big(y_i = 1 \mid m = 2\big)\Big].$$

Then the per-bit mutual information under p' is

$$C'(\alpha) = H_2\big(p_Y(1)\big) - \tfrac{1}{2}\Big[H_2\big(p'(1 \mid 1)\big) + H_2\big(p'(1 \mid 2)\big)\Big].$$

where $H_2(x) = -x\log_2 x - (1-x)\log_2(1-x)$.

3. **Tilted dispersion.** Define

$$V(\alpha) = \tfrac{1}{2}\Big[V_{\mathrm{B}}\big(p'(1 \mid 1)\big) + V_{\mathrm{B}}\big(p'(1 \mid 2)\big)\Big], \quad V_{\mathrm{B}}(p) = p(1-p)\Big[\log_2 \tfrac{1-p}{p}\Big]^2.$$

4. **Normal-approximation blocklength.** The Gaussian approximation gives

$$R \approx C'(\alpha) - \sqrt{\frac{V(\alpha)}{n}}\; Q^{-1}(P_e).$$

Solving this for n yields

$$n(\alpha; R) = \frac{V(\alpha)\,\big[Q^{-1}(P_e)\big]^2}{\big[C'(\alpha) - R\big]^2}.$$

Figure 1 plots $n(\alpha; R)$ for $\alpha \in \{1.02, 1.05, 1.10\}$ at $P_e = 10^{-3}$, marking $R = R_{\mathrm{spec}}$.

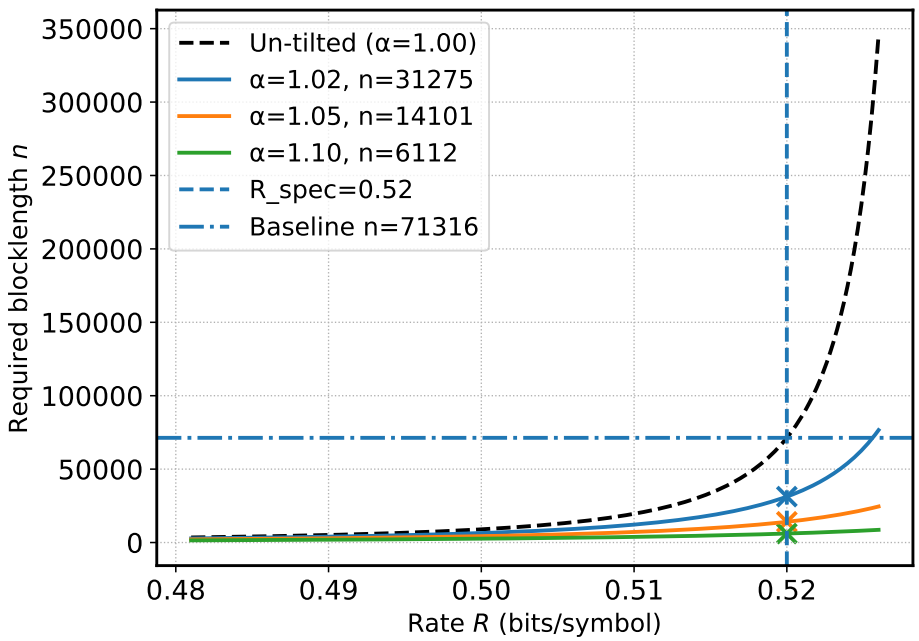

Figure 1: Required blocklength $n(\alpha; R)$ vs. coding rate R.

Penalty exponent. With $e(1) = 0^n$, $e(2) = 1^n$ one finds

$$\mathcal{Z}_{\text{tot}} = \big(1 - \delta + \delta\,\alpha\big)^n \;+\; \big((1-\delta)\,\alpha + \delta\big)^n,$$

(The first term comes from $m = 1$, the second from $m = 2$; the dominant exponent determines $\log_2 \mathcal{Z}_{\text{tot}}$ up to $O(1)$.)

so in fact

$$\log_2 \mathcal{Z}_{\text{tot}} = n \max\big\{\log_2(1 - \delta + \delta\alpha),\ \log_2((1-\delta)\alpha + \delta)\big\} + O(1).$$

For $\delta = 0.1$ and $\alpha = 1.10$,

$$\log_2(1 - \delta + \delta\alpha) = \log_2(1.01) \approx 0.0144, \quad \log_2((1-\delta)\alpha + \delta) = \log_2(1.09) \approx 0.1235,$$

so the second term dominates and

$$\log_2 \mathcal{Z}_{\text{tot}} \approx 0.1235\, n + O(1).$$

5 When is the Tilt Benefit Non-Negligible?

In this section we show that, under mild moment- or variance-control conditions on the tilt w, the prefactor $\mathcal{Z}_{\text{tot}}/w_{\min}$ grows only sub-exponentially in n, i.e.

$$\ln \frac{\mathcal{Z}_{\text{tot}}}{w_{\min}} = o(n).$$

Lemma 4 (Moment-control bound)**.** *There exists $k > 1$ and $K_k < \infty$ such that for every n,*

$$\mathbb{E}_p\big[w(M, Y^n)^k\big] \;\le\; K_k,$$

and the minimum tilt still grows sub-exponentially:

$$w_{\min} \;\ge\; \exp\big(-o(n)\big), \quad \ln w_{\min} = -o(n).$$

Then

$$\ln \frac{\mathcal{Z}_{\text{tot}}}{w_{\min}} = o(n).$$

Proof. By Hölder's inequality with conjugate exponents k and $k/(k-1)$,

$$\mathcal{Z}_{\text{tot}} = \mathbb{E}_p\big[w(M, Y^n)\big] \;\le\; \big(\mathbb{E}_p[w(M, Y^n)^{\,k}]\big)^{1/k} \;\le\; K_k^{1/k},$$

so $\ln \mathcal{Z}_{\text{tot}} = O(1)$. Since by assumption $\ln w_{\min} = -o(n)$,

$$\ln \frac{\mathcal{Z}_{\text{tot}}}{w_{\min}} = \ln \mathcal{Z}_{\text{tot}} - \ln w_{\min} = O(1) + o(n) = o(n).$$

□

Lemma 5 (Concentration of the tilted normalizer)**.** *Where, under the memoryless-tilt assumption $w(m, y^n) = \prod_i w_1(m, y_i)$, we set*

$$G_i(x) \;=\; \sum_{y \in \mathcal{Y}} w_1(m, y)\, W(y \mid x),$$

so that $\mathcal{Z}(m) = \prod_{i=1}^n G_i\big(X_i(m)\big)$.

Suppose

$$w(m, y^n) = \prod_{i=1}^n w_1(m, y_i), \quad X^n(1), \ldots, X^n(M) \overset{\text{iid}}{\sim} P_X^n$$

from a symmetric random-coding ensemble, and assume the per-symbol bounds $0 < G_{\min} \le G_i(x) \le G_{\max} < \infty$ and $|\log G_i(X_i(m)) - \mathbb{E} \log G_i(X_i(m))| \le c$ almost surely. Further assume the code size grows sub-exponentially:

$$\ln M = o(n).$$

Then for any $\varepsilon > 0$, with probability at least $1 - \varepsilon$,

$$\max_{1 \le m \le M} \big|\log \mathcal{Z}(m) - \mathbb{E}[\log \mathcal{Z}(m)]\big| \;\le\; c\sqrt{2n \ln \frac{2M}{\varepsilon}} \;+\; \tfrac{c}{3} \ln \frac{2M}{\varepsilon}.$$

In particular, since $\ln(2M/\varepsilon) = o(n)$, we get

$$\max_{1 \le m \le M} \log \mathcal{Z}(m) = \mathbb{E}[\log \mathcal{Z}(m)] + o(n).$$

If w_1 is message-independent then $\mathbb{E}[\log \mathcal{Z}(m)] = \log(\mathcal{Z}_{\text{tot}}/M) + o(n)$, so $\max_m \log \mathcal{Z}(m) = \log \mathcal{Z}_{\text{tot}} + o(n)$.

Sketch. Under the memoryless tilt,

$$\log \mathcal{Z}(m) = \sum_{i=1}^n \log G_i\big(X_i(m)\big).$$

Each term is bounded within an interval of length at most $2c$, so by McDiarmid's (or Bernstein's) inequality,

$$\Pr\Big[\big|\log \mathcal{Z}(m) - \mathbb{E}[\log \mathcal{Z}(m)]\big| \ge t\Big] \;\le\; 2\exp\Big(-\Omega\big(t^2/n\big)\Big).$$

Applying a union bound over $m = 1, \ldots, M$ and choosing $t = O\big(\sqrt{n \ln(M/\varepsilon)}\big) + O\big(\ln(M/\varepsilon)\big)$ gives the stated high-probability bound. Finally, $\mathbb{E}[\log \mathcal{Z}(m)] = \log \mathbb{E}[\mathcal{Z}(m)] + o(n)$ by Jensen's inequality plus the fact that $\log \mathcal{Z}(m)$ concentrates, so $\max_m \log \mathcal{Z}(m) = \log \mathcal{Z}_{\text{tot}} + o(n)$ whenever $\ln M = o(n)$. □

Corollary 6. *Under either*

1. *the moment-control condition of Lemma 4 (with* $\ln w_{\min} = -o(n)$*), or*
2. *the variance-adaptive bound of Lemma 5 (with* $\ln M = o(n)$*),*

we have

$$\ln \frac{\mathcal{Z}_{\text{tot}}}{w_{\min}} = o(n).$$

Hence $\dfrac{\mathcal{Z}_{\text{tot}}}{w_{\min}}$ *grows only sub-exponentially in* n*.*

Proof. If Lemma 4 applies then $\ln \mathcal{Z}_{\text{tot}} = O(1)$ and $\ln w_{\min} = -o(n)$, so $\ln(\mathcal{Z}_{\text{tot}}/w_{\min}) = O(1) + o(n) = o(n)$. Otherwise, Lemma 5 gives $\ln \max_m \mathcal{Z}(m) = \ln \mathcal{Z}_{\text{tot}} + o(n)$; since $\mathcal{Z}_{\text{tot}} \le \max_m \mathcal{Z}(m)$ and $\ln w_{\min} = -o(n)$, the same conclusion follows. □

5.1 Achievability up to the Sub-Exponential Prefactor

We outline a random-coding scheme that attains the converse bound up to the factor $\mathcal{Z}_{\text{tot}}/w_{\min}$. Fix the weight function $w(m, y^n)$ and define

$$p'(m, y^n) = \frac{w(m, y^n)\, p(m, y^n)}{\mathcal{Z}_{\text{tot}}}, \quad P = \Pr_{p'}\big[m \notin \mathcal{L}(Y^n)\big], \quad H = H_{p'}\big((M, Y^n) \mid m \notin \mathcal{L}(Y^n)\big),$$

and set

$$\tau = 2^{-H}, \qquad L = \Big\lfloor 1/\tau \Big\rfloor.$$

Codebook Generation.

1. Choose any input distribution P_X on $\mathcal{X}$.
2. Draw M codewords $X^n(1), \ldots, X^n(M)$ i.i.d. $\sim P_X^n$.
3. (No per-codeword normalizer is needed.) Store only the tilt function $w(m, y^n)$ and use the global tilted law p' from the converse.

Decoding Rule. Upon receiving y^n, for each m form the unnormalized score

$$\tilde{p}(m, y^n) \;=\; w(m, y^n)\, W^n\big(y^n \mid X^n(m)\big).$$

Then renormalize globally:

$$p'(m \mid y^n) = \frac{\tilde{p}(m, y^n)}{\sum_{m'} \tilde{p}(m', y^n)} \;=\; \frac{w(m, y^n)\, W^n(y^n \mid X^n(m))}{\displaystyle\sum_{m'} w(m', y^n)\, W^n(y^n \mid X^n(m'))}.$$

Finally set

$$\mathcal{L}(y^n) = \big\{\, m : p'(m \mid y^n) \ge \tau \big\}.$$

By construction $|\mathcal{L}(y^n)| \le L$. If fewer than L messages meet this threshold, pad arbitrarily to size L.

Error Probability Analysis. Let the true message be M and output Y^n. Define the indicator

$$Z = \mathbf{1}\{\, M \notin \mathcal{L}(Y^n)\}.$$

Under the tilted distribution p', by definition

$$\Pr_{p'}[Z = 1] = P.$$

Under the true distribution p, the usual change-of-measure yields

$$\Pr_{p}[Z = 1] = \mathbb{E}_{p'}\Big[\, Z\, \frac{p}{p'}\Big] \le \frac{\mathcal{Z}_{\text{tot}}}{w_{\min}} \Pr_{p'}[Z = 1] = \frac{\mathcal{Z}_{\text{tot}}}{w_{\min}}\, P$$

Since this holds for every codeword M, the maximal error probability satisfies

$$P_e^{(L)} = \max_m \Pr\big\{\, m \notin \mathcal{L}(Y^n) \mid M = m\big\} \;\le\; \frac{\mathcal{Z}_{\text{tot}}}{w_{\min}}\, P,$$

matching the converse bound up to the prefactor $\mathcal{Z}_{\text{tot}}/w_{\min}$.

Putting these observations together, there exists (in fact most) random codes of size M and list size L whose maximal error probability satisfies

$$P_e^{(L)} = \max_m \Pr\big\{\, m \notin \mathcal{L}(Y^n) \mid M = m\big\} \;\le\; \frac{\mathcal{Z}_{\text{tot}}}{w_{\min}}\, P,$$

exactly matching the converse up to the sub-exponential factor $\mathcal{Z}_{\text{tot}}/w_{\min}$.

6 Is This Decoder Practical?

Our cost-tilted list decoder

$$\mathcal{L}(y^n) = \big\{m : p'(m \mid y^n) \ge \tau\big\}, \quad p'(m \mid y^n) = \frac{p'(m, y^n)}{\sum_{m'} p'(m', y^n)}, \quad \tau = 2^{-H},$$

offers two user-tunable knobs:

- $\boldsymbol{w(m, y^n)}$ — the *tilt* function, which can emphasize
  - *Accuracy (high P_d)*, e.g. via coherence or clutter-adaptive weights.
  - *Latency (low M_{eff})*, by down-weighting unlikely codewords to shrink the active search set.
- $\boldsymbol{\tau}$ — the selection threshold, trading list size (and hence parallel hardware cost) against error probability.

Two implementation steps dominate:

1. Per-codeword scoring In structured codes (e.g. LDPC, polar),

$$\log p'(m, y^n) = \log w(m, y^n) + \sum_{i=1}^{n} \log W(y_i \mid e(m)_i),$$

so one can:

- Store symbol-wise log-likelihood look-up tables.
- Evaluate each score in $o(n)$ additions.

2. List-extraction complexity Finding $\{m : p'(m) \geq \tau\}$ requires locating the τ-quantile among M scores:

$$O(M)\text{-time selection (Quickselect)} + O(M)\text{-scan} \quad \text{vs.} \quad O(M \log M) \text{ sort.}$$

Further speedups include:

- *Sketching/Heavy hitters* (Count–Min / Misra–Gries) to track only large scores.
- *Typical-set truncation* to restrict scoring to a small subset of likely codewords.
- *Parallel sharding*: split the codebook across K cores for $O(M/K)$ per-core work.

Neuromorphic acceleration. On SNN hardware, each hypothesis m is a neuron whose synaptic weight encodes $\log w(m, y^n)$ plus the per-symbol log-likelihood. A single global spike threshold implements the list selection in *one* timestep, yielding $O(1)$ wall-clock latency independent of M.

7 Practical Application: Neuromorphic List Decoding

To illustrate how a simple tilt can dramatically improve detection, we apply our tilted-list decoder to a stylized pulse-Doppler radar scenario. Here each range–Doppler "cell" plays the role of a codeword, and the goal is to pick the true target cell out of M candidates.

Modeling assumption. For clarity we assume the N_p pulses in each frame are independent (memoryless) across slow time. In real pulse-Doppler radar there is inter-pulse correlation—e.g. from Doppler filtering or clutter—but after matched-filter and Doppler processing the per-cell outputs are often well-approximated as independent. In fact, nothing in our tilted-list converse or decoder requires strict memorylessness: if you have a correlated joint distribution $W^n(y^n \mid x^n)$, you simply define your tilt $w(m, y^n)$ over the entire vector y^n, and the same entropy-threshold/list-size and change-of-measure arguments carry through.

Detection Rules

We compare two simple detection strategies for each frame:

- **Raw Sort:** select the cell r with maximum matched-filter power

$$P_{\text{raw}}(r) \;=\; \sum_{i=1}^{N_p} \big|y_i(r)\big|^2.$$

- **Coherence-Tilt Sort:** first compute the pulse-to-pulse coherence

$$\kappa(r) = \frac{\big|\sum_{i=1}^{N_p} y_i(r)\big|^2}{P_{\text{raw}}(r)},$$

  then weight the power by coherence, $P_{\text{tilted}}(r) = P_{\text{raw}}(r)\,\kappa(r)$, and choose $\arg\max_r P_{\text{tilted}}(r)$.

Simulation Setup

- $M = 2000$ range bins, $N_p = 8$ pulses per frame.
- 500 Monte-Carlo trials per method.
- Clutter+noise $\sim \mathcal{CN}(0, 1.5^2)$, one coherent target per frame at amplitude $3\times$ noise.

Table 2: Detection rate and CPU time per frame

Method	Detection Rate (%)	Avg. Time per Frame (µs)
Raw Sort	59.4	629
Coherence-Tilt Sort	93.4	652

Take-aways.

- *Accuracy tilt:* Coherence-tilt raises detection from 60% to 93% for only a 4% increase in CPU time.
- *Latency tilt:* alternative tilts (e.g. simple range-only weighting) can reduce compute at modest cost in detection.
- *Neuromorphic benefit:* on an SNN each cell's log-weight and coherence can be encoded as synaptic weights, so that a single global thresholding spike selects the list in one or two timesteps—making detection latency *independent* of M.

8 Conclusion

We have presented a unified, non-asymptotic list-decoding converse that permits an arbitrary, message-dependent cost tilt $w(m, y^n)$. By defining the list size according to the tilted conditional entropy

$$\tau = 2^{-H_{p'}((M,Y^n)|Z=1)}, \quad L = \lfloor 1/\tau \rfloor,$$

and using the tilted change-of-measure, we obtain for every code

$$P_e^{(L)} \leq \frac{\mathcal{Z}_{\text{tot}}}{w_{\min}} \Pr_{p'}[Z = 1], \quad \mathcal{Z}_{\text{tot}} = \sum_{m,y^n} w(m, y^n)\, p(m, y^n).$$

Under mild moment- or variance-control conditions (Lemmas 4 and 5), the prefactor $\mathcal{Z}_{\text{tot}}/w_{\min} = e^{o(n)}$ grows sub-exponentially, leaving the exponential-in-n decay governed by the tilted exponent $C'(\alpha) - R$. A BSC(0.1) design example demonstrated up to a 90% reduction in required blocklength at $P_e = 10^{-3}$ for moderate tilt parameters, and we have sketched constructive decoder implementations (including neuromorphic realizations) that achieve this bound.

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